Sangeon Jeon

Ph.D. Student, Electrical and Computer Engineering, Georgia Institute of Technology
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Education

Georgia Institute of Technology Jan 2026–present
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Xiaoli Ma GPA: 4.00/4.0

Georgia Institute of Technology Jan–Dec 2025
M.S. in Electrical and Computer Engineering
Advisor: Prof. Xiaoli Ma GPA: 4.00/4.0

KAIST Feb 2018–Feb 2025
B.S. in Electrical Engineering
Advisor: Prof. Jinseok Choi GPA: 3.95/4.3

Georgia Institute of Technology Jan 2023–Dec 2024
B.S. in Electrical and Computer Engineering
GPA: 4.00/4.0
Bachelor's degrees completed through the KAIST–Georgia Tech Dual B.S. Degree Program.

Research Interests

Next-generation waveform design and synchronization, especially orthogonal chirp division multiplexing (OCDM); TI-ADC mismatch calibration; Bayesian statistics.

Research Experience

Next-Generation Waveforms and Synchronization Current
Georgia Institute of Technology; advisor: Prof. Xiaoli Ma

- Research on wide-range carrier frequency offset estimation for OCDM using a single pilot block in multipath channels, with an emphasis on identifiability and model-based estimation.
- Related first-author manuscript under review at ICASSP 2027.

Joint Channel and TI-ADC Mismatch Estimation Sep 2024–Oct 2026
OFDM receiver research

- Investigated joint wireless-channel and time-interleaved ADC mismatch estimation, emphasizing reliability and computational efficiency; first-author manuscript under review at ICC 2027.

Bayesian Modeling and Meta-Analysis
Collaboration with the Center for Digital Health, Kyung Hee University
Co-first author of three published population-health modeling studies and a hemophilia manuscript. The hemophilia study also involves the World Federation of Hemophilia.

CLOC Unnatural Subspace System Identification Spring 2024
SIPLAB, Georgia Institute of Technology; Christopher J. Rozell
Worked on a computational testbench for optical control of neural populations, including recurrent neural-network and opsin-dynamics modeling.

Undergraduate Research Assistant Summer 2020
NICA Lab, KAIST
Developed an approach to neuron segmentation in microscopy images and studied convolutional neural networks.

Publications

† Co-first author. ‡ Equal contribution. Author names appear in manuscript order.

Conference Submissions

Wide Range Carrier Frequency Offset Estimation for OCDM With a Single Pilot
Sangeon Jeon, Xiaoli Ma
ICASSP 2027. Under review.

Robust and Computationally Efficient Joint Channel and TI-ADC Mismatch Estimation for OFDM
Sangeon Jeon, Jiwon Sung, Xiaoli Ma, Jinseok Choi
ICC 2027. Under review.

Journal Articles

Global Burden of Age-Related Macular Degeneration, with Forecast up to 2050: A Modeling Study
Yi Deun Jeong†, **Sangeon Jeon**†, Wonseok Shin†, Seoyoung Park†, Taehwan Park, Tae Hyeon Kim, Lee Smith, Masoud Rahmati, Jee Myung Yang, Dong Keon Yon
Ophthalmology Retina, 2026.

Global Prevalence of Obsessive-Compulsive and Related Disorders: A Systematic Review and Modeling Study
Yi Deun Jeong†, Yejun Son†, **Sangeon Jeon**†, Hanseul Cho, Seung Woo Ryuk, Yeona Jo, Guillaume Fond, Laurent Boyer, Lee Smith, Samuele Cortese, Paolo Fusar-Poli, Dong Keon Yon, Marco Solmi
American Journal of Psychiatry, 2026.

Global Prevalence of Spinal Muscular Atrophy in 204 Countries and Territories, 1960-2024: A Systematic Review and Bayesian Modeling Study
Yi Deun Jeong†, Yejun Son†, **Sangeon Jeon**†, Kanghyun Lee, Hyeon Jin Kim, Yeongtaek Hwang, José Francisco López-Gil, Masoud Rahmati, Ho Geol Woo, Jiseung Kang‡, Dong Keon Yon‡
Pediatric Neurology, 2026.

Manuscript

Global disparities and unmet treatment needs in hemophilia toward 2030
Yi Deun Jeong†, **Sangeon Jeon**†, Yejun Son†, Soeun Kim†, Ellia Tootoonchian, Donna Coffin, Taehwan Park, Dong Keon Yon, Jeffrey Stonebraker, Glenn F. Pierce
Manuscript.

Awards and Honors

Roger Webb Award, ECE Senior Scholar Awards — Spring 2025
Georgia Institute of Technology

Grand Prize (1st Place), KAIST Makerathon — Summer 2024
When to Eat team leader; KRW 2,500,000 team award

Dean's List, Georgia Institute of Technology — 2023–2024
Spring 2023, Fall 2023, Spring 2024

Dean's List, KAIST — 2018 / 2022
Spring 2018, Fall 2022

Industry Experience

Software Engineer Intern Jun–Aug 2023
Dong Il Vision, Anyang, South Korea

- Developed a C# WPF application for annotating object locations in images for a machine-learning computer-vision pipeline.

Human Resources Intern Jun–Aug 2018
Dong Il Technology LTD, Hwaseong, South Korea

- Automated employee performance-evaluation sheet generation and distribution using Excel VBA.

Selected Engineering Projects

Robotic Tic-Tac-Toe Fall 2024
KAIST; C++, Eigen, DYNAMIXEL
Project combined rule-based game logic, keyboard input, hand-taught waypoints, joint interpolation, and Jacobian-based pressing adjustment on a four-degree-of-freedom robotic arm.

When to Eat: RFID Food Storage Reminder Summer 2024
KAIST Makerathon; four-person team leader
Led a team developing reusable RFID tokens and a C#/XAML Windows UWP elapsed-time reminder prototype. Received the Grand Prize (1st place).

Venus Flytrap Bioelectrical Recording and Playback Spring 2024
Georgia Institute of Technology; Senior Design
Designed and programmed the complete digital system: Seeeduino XIAO firmware, MATLAB/App Designer interface, USB/serial communication, monitoring, recording, waveform upload, and playback. Team demonstrated electrically stimulated trap closure.

RoBACHtic Symphony: Educational Robotic Violin Kit Fall 2023
Georgia Institute of Technology; mechanical design and fabrication lead
Used SolidWorks, 3D printing, and laser cutting for mechanical design and integration revisions. Team prototype combined bow/string-position motion with micro:bit electronic tones.

Teaching and Mentoring

ECE Tutor Spring 2024
Georgia Institute of Technology
Tutored signal processing, circuit analysis, digital system design, and engineering software design.

Hive Peer Instructor Spring 2024
The Hive Makerspace, Georgia Institute of Technology
Supported laser-cutter and 3D-printer users; led a hometown-map fabrication workshop.

Introduction to Programming Teaching Assistant Spring 2020
KAIST
Supported programming labs and graded assignments.

Programming and Calculus Tutor 2019
KAIST
Tutored Introduction to Programming in spring and Calculus I in fall.

Technical Skills and Languages

Programming: MATLAB; Python (PyMC, NumPy, SciPy, pandas, PyTorch); C# (WPF/UWP); C/C++; Java; Excel VBA.
Design and laboratory: SolidWorks; HFSS; NI Multisim; 3D printing; laser cutting; oscilloscopes, function generators, and multimeters.
Languages: Korean (native); English (full professional proficiency); Japanese (conversational); Chinese (beginner).